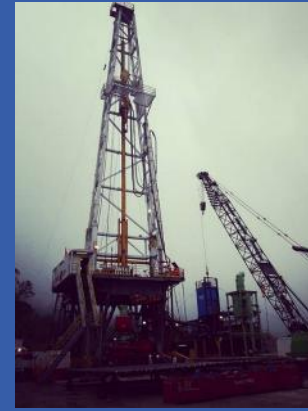




Introduction of Geothermal Energy, Exploration. Development, Monitoring

“Jakarta Drilling Society Class Batch#6”

Mauliate Agustinus Hamonangan Sihotang M.T

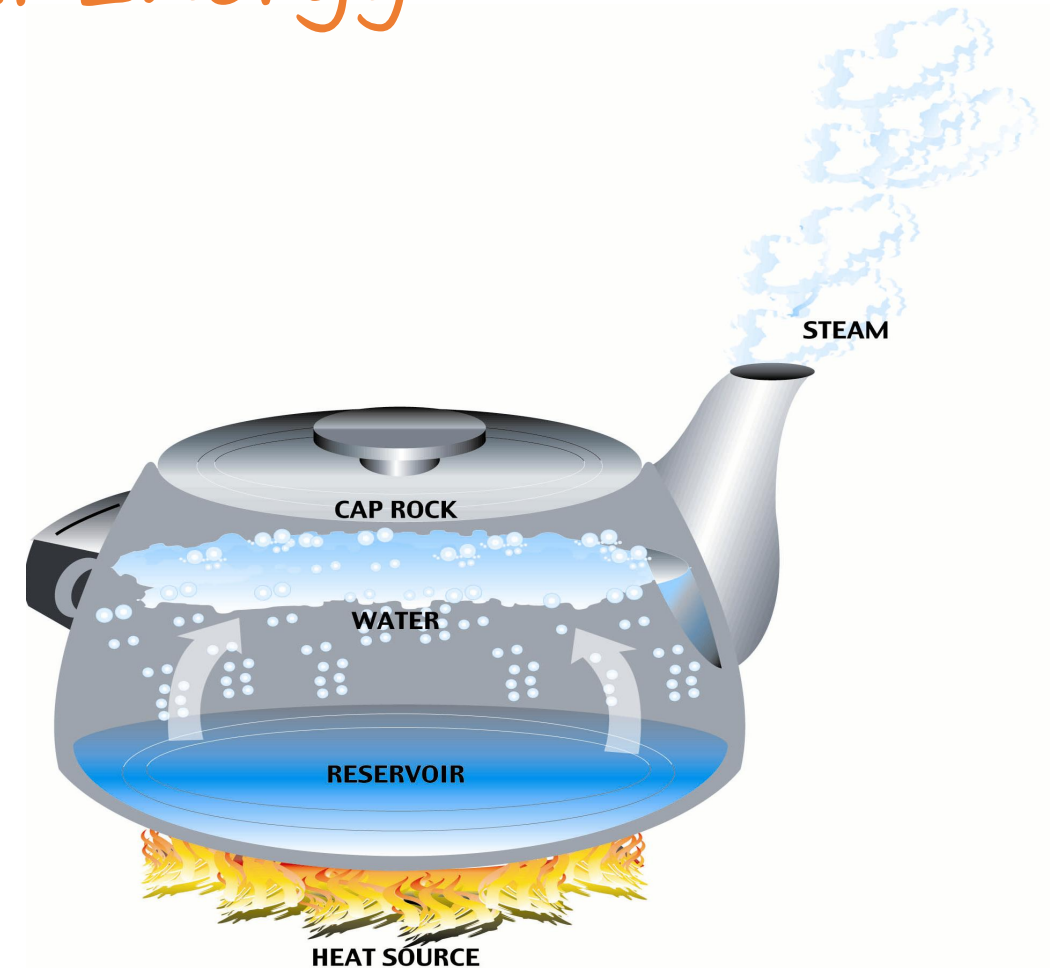
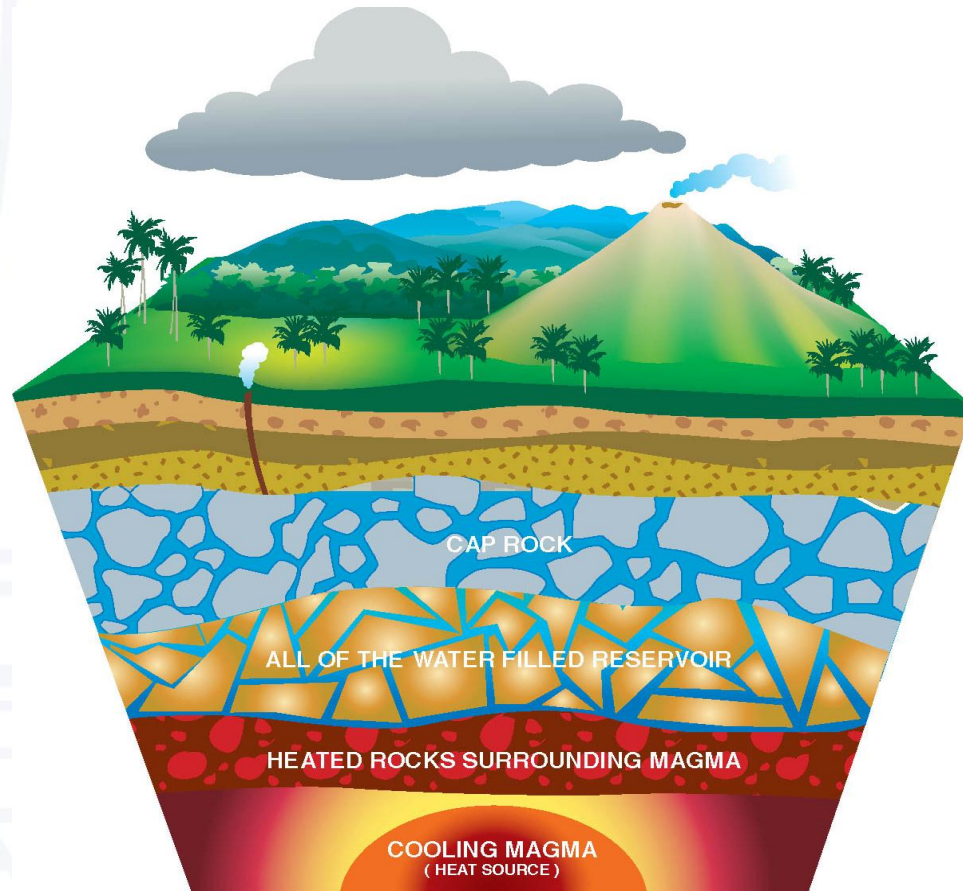


- ▶ Renewable Energy Enthusiast
- ▶ Current Position : Manager of New Opportunities (Exploration) and Technical Support PT.SUPREME ENERGY
- ▶ Education :
 - ▶ M.Eng on Geothermal Engineering ITB
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- ▶ Past Experience :
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 - ▶ Sr.Development Geologist PT.SUPREME ENERGY
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 - ▶ Sr.Exploration Geologist PT.Sorik Merapi Geothermal Power – OTP Geothermal
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Outlines

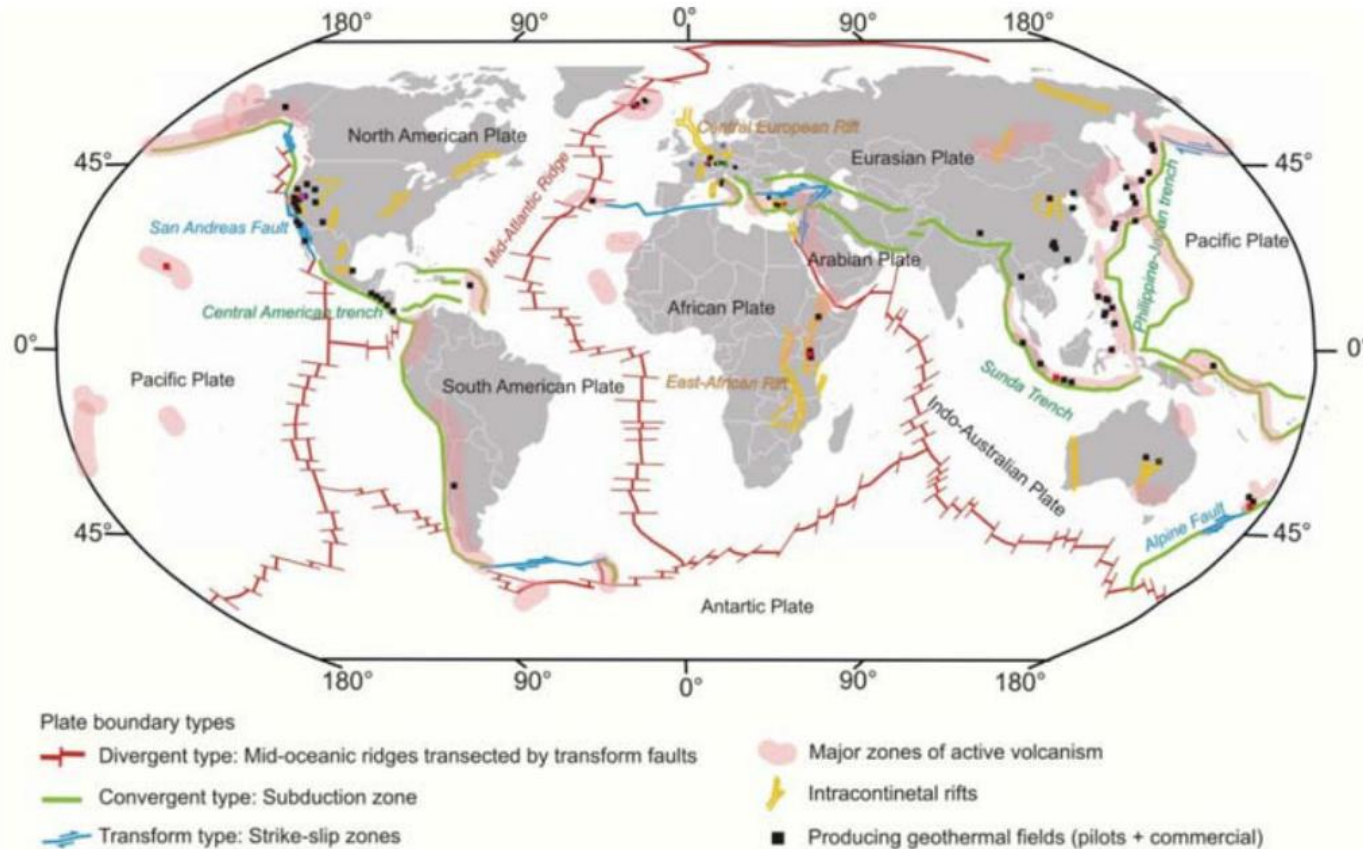
- Introduction to Geothermal Energy : Exploration, Development, and Reservoir Management
- Case Studies

What is Geothermal Energy



The elements of a geothermal system works like boiling water in a kettle.

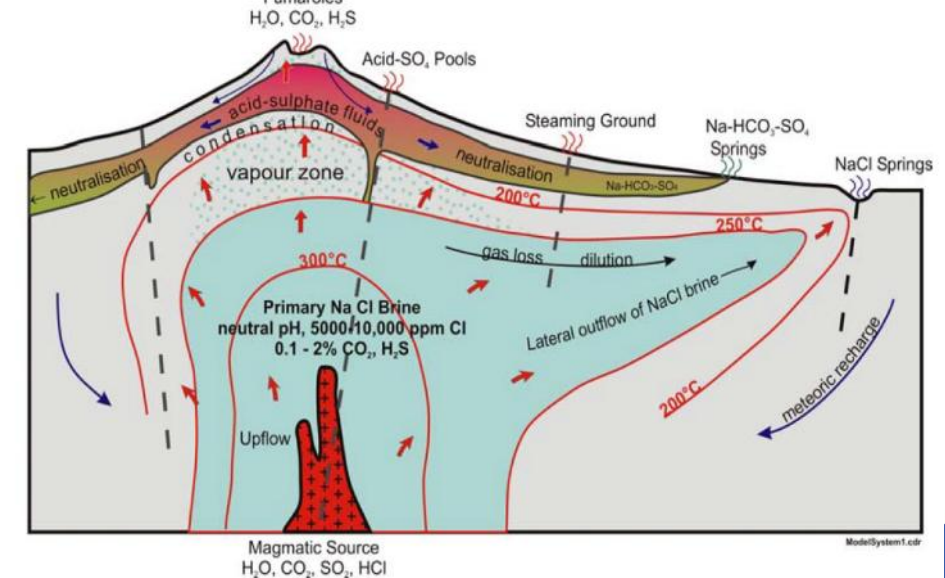
Where to find geothermal energy?



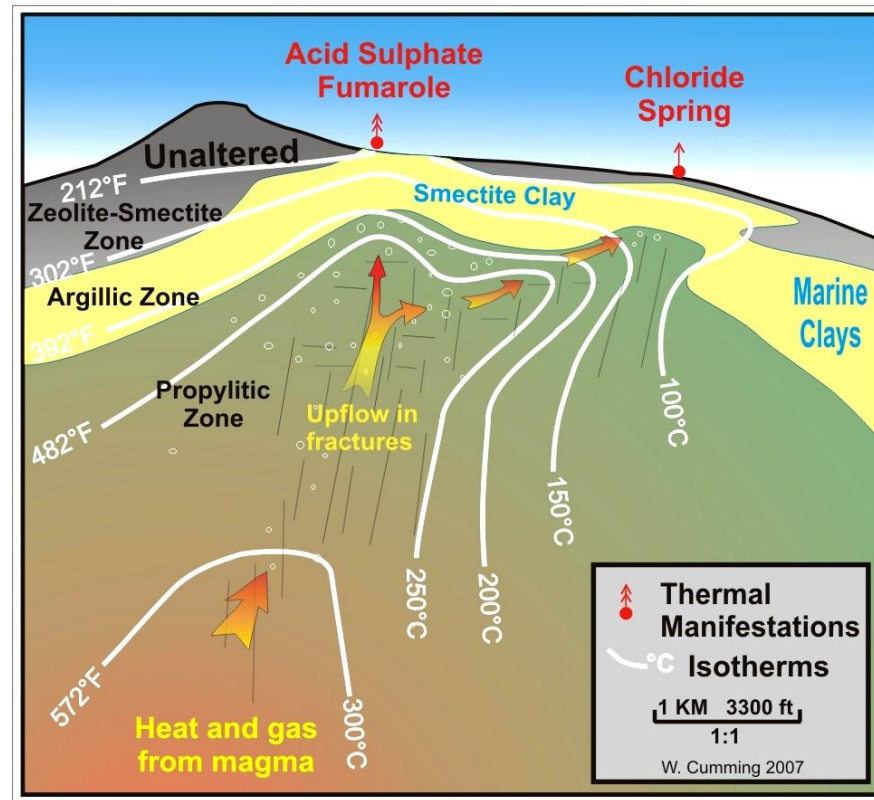
Gehring and Loksha, 2012

Geothermal Potency in Indonesia 23.77
GW VS Installed Capacity 2.28 GW
(ESDM, 2021)

- ✓ Most of geothermal related with volcanism on plate boundary
- ✓ Some of geothermal system also related with major structural geology features like faults



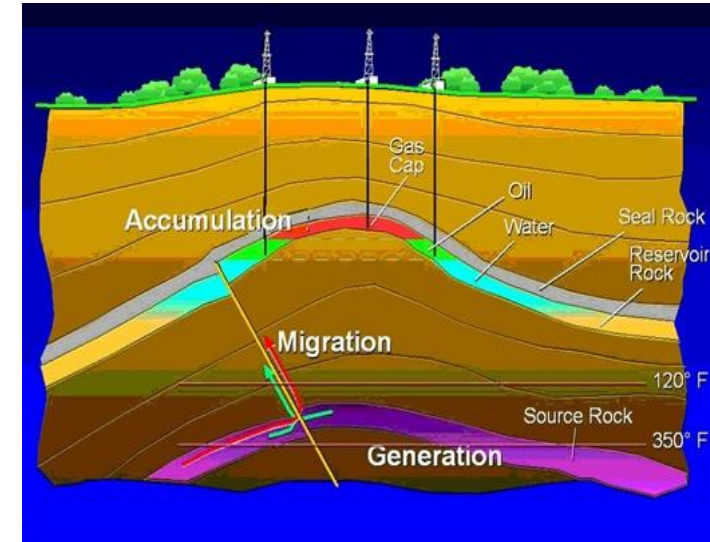
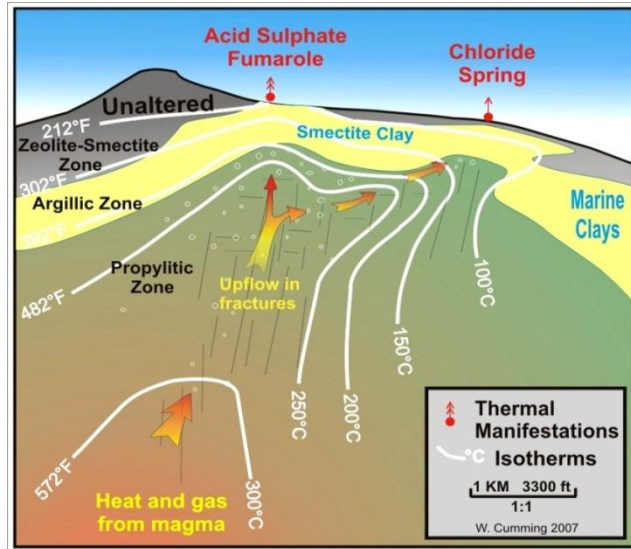
Geothermal system



Cumming, 2009

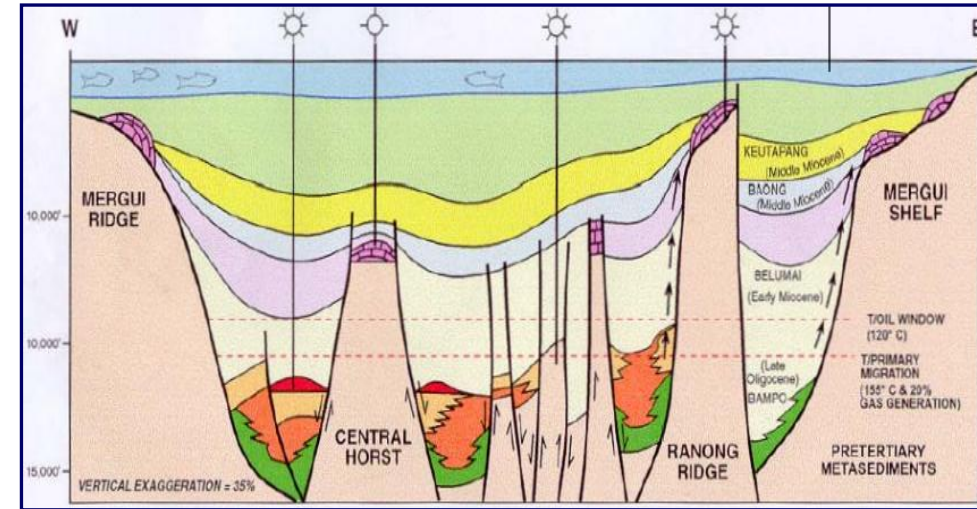
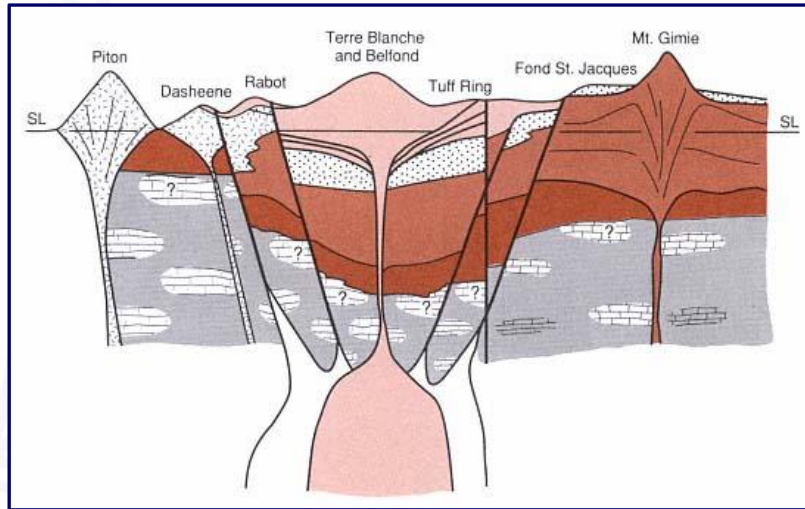
- *Heat source* → Heat flow
- *Reservoir and fluid pathway* → Porosity and permeability
- *Cap rock (clay cap)* → Controlling factor
- *Circulating fluid* → Temperature and chemistry

Geothermal vs. Petroleum system



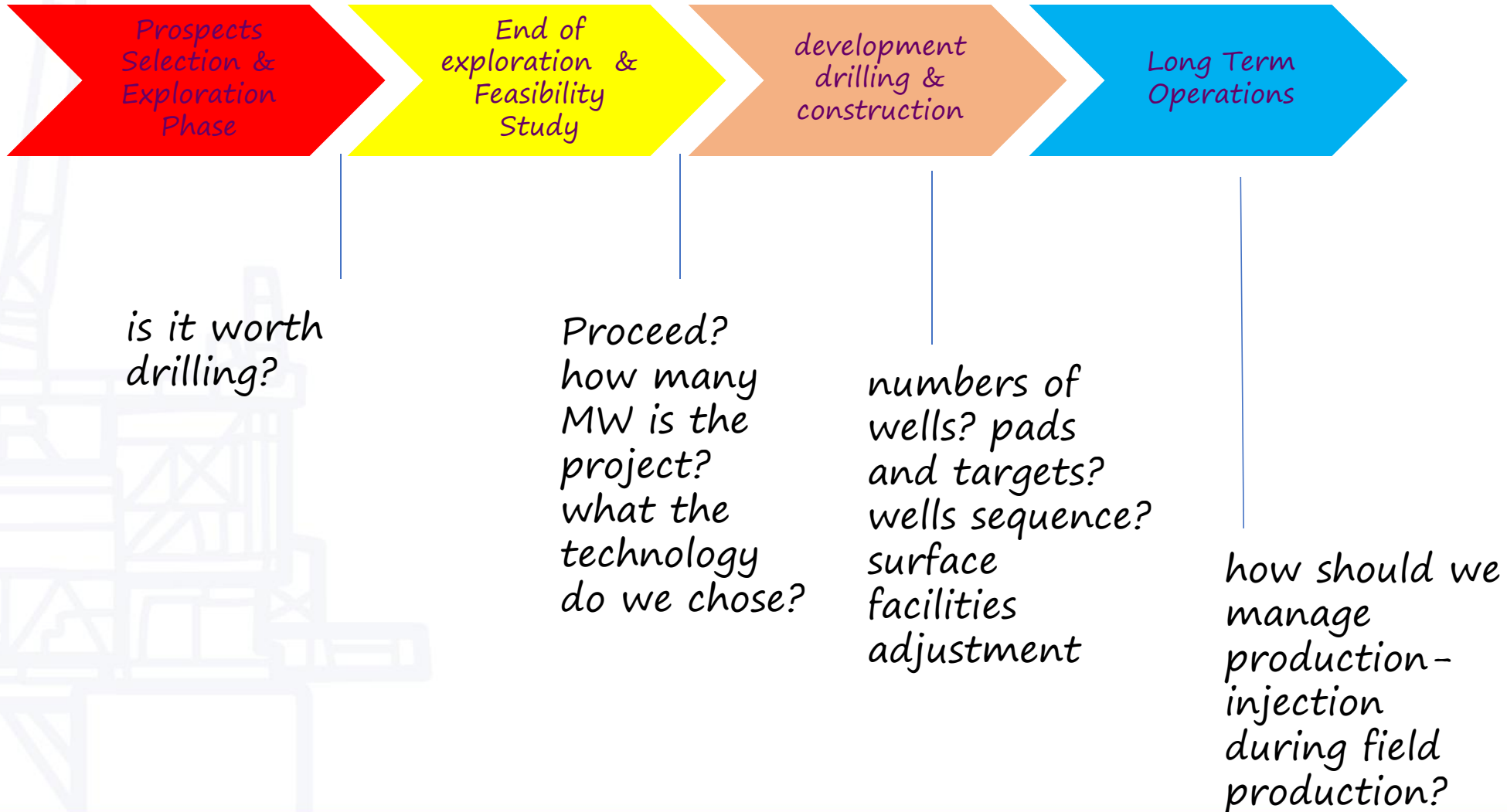
Geothermal	Component	Petroleum
Heat source is cooling intrusion or batholith not too old (Quaternary) to provide sufficient heat	Source	Source rock is organic rich strata mature enough to generate hydrocarbon
Commonly within volcanic environment	Reservoir	Commonly sedimentary environment
Commonly clay alteration zone (smectite layer) Protecting hot system from surrounding cold water	Cap	Impermeable strata (shale, salt, etc.) Trapping hydrocarbon and prevent further migration
Circulating hydrothermal fluid	Fluid	Accumulated hydrocarbon (oil or gas)
Commonly under pressured	Pressure	Commonly over pressured

Geothermal vs Petroleum system

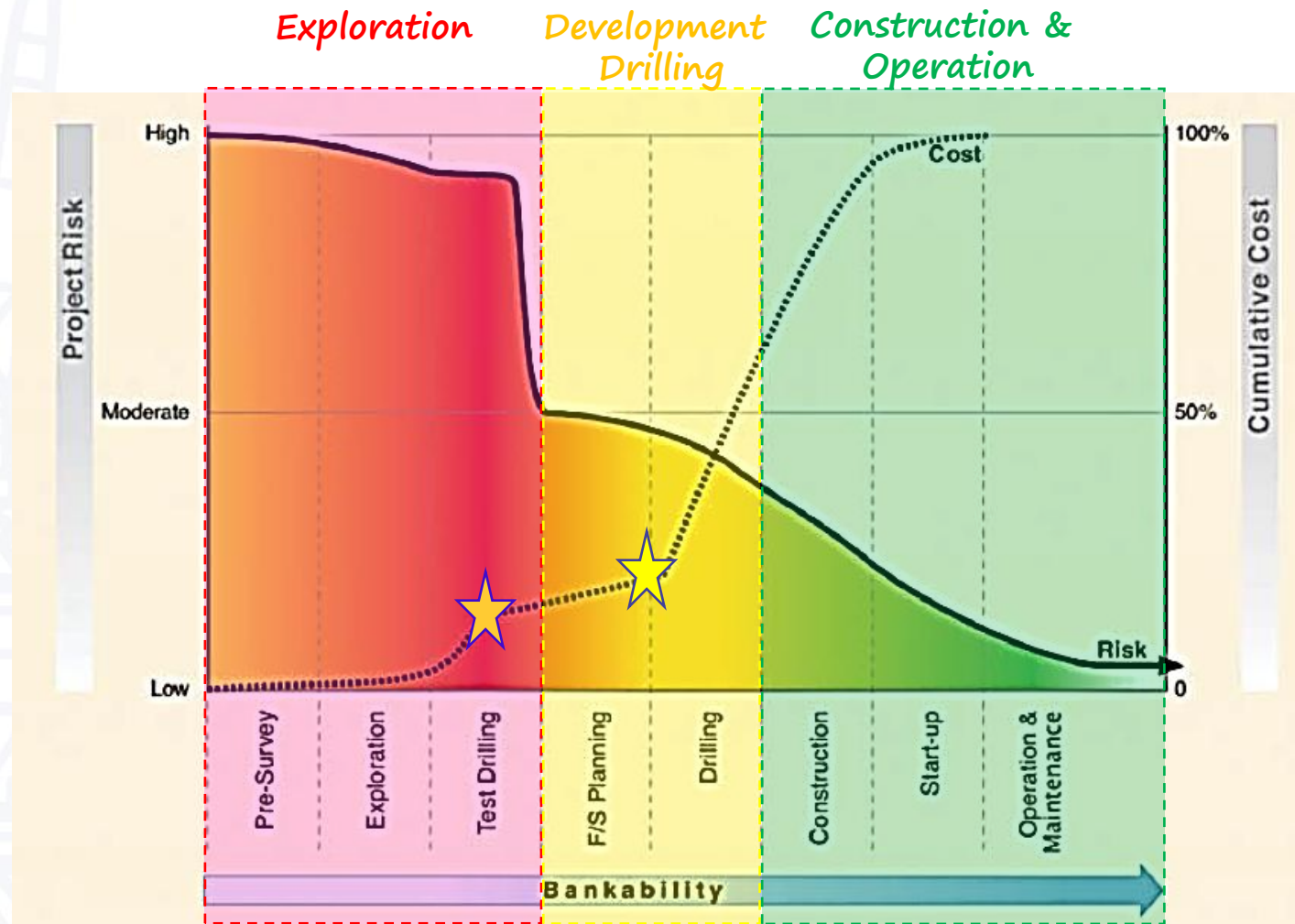


Geothermal	Component	Petroleum
Covers local geological/volcanic setting	Spatial	Regional/basin-wide setting
Quaternary volcanoes overlying the older formation Volcano-stratigraphy	Temporal	Wider time frame Basin evolution over various tectonic events Lithostratigraphy, tectono-stratigraphy, sequence-stratigraphy, etc.

Key stages in geothermal development



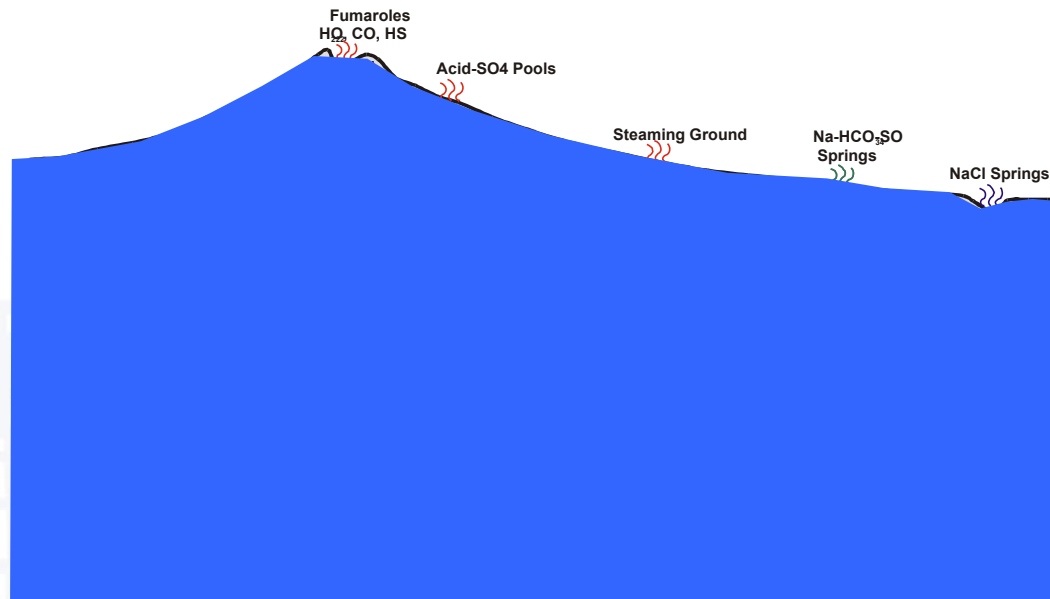
Geothermal Development (Stages)



Gehring and Loksha,
2012.

- ✓ Stage by stage activities in developing geothermal field is intended to gradually reduce the risks
- ✓ The **highest risk** is related to subsurface activity (upstream), i.e. pre-feasibility and exploration stage
- ✓ The **highest cost** is related to surface facility construction
- ✓ The risk is significantly reduced after exploration drilling

Is it worth Drilling?



Key Questions

- How hot?
- How permeable?
- Is fluid benign?
- How big?

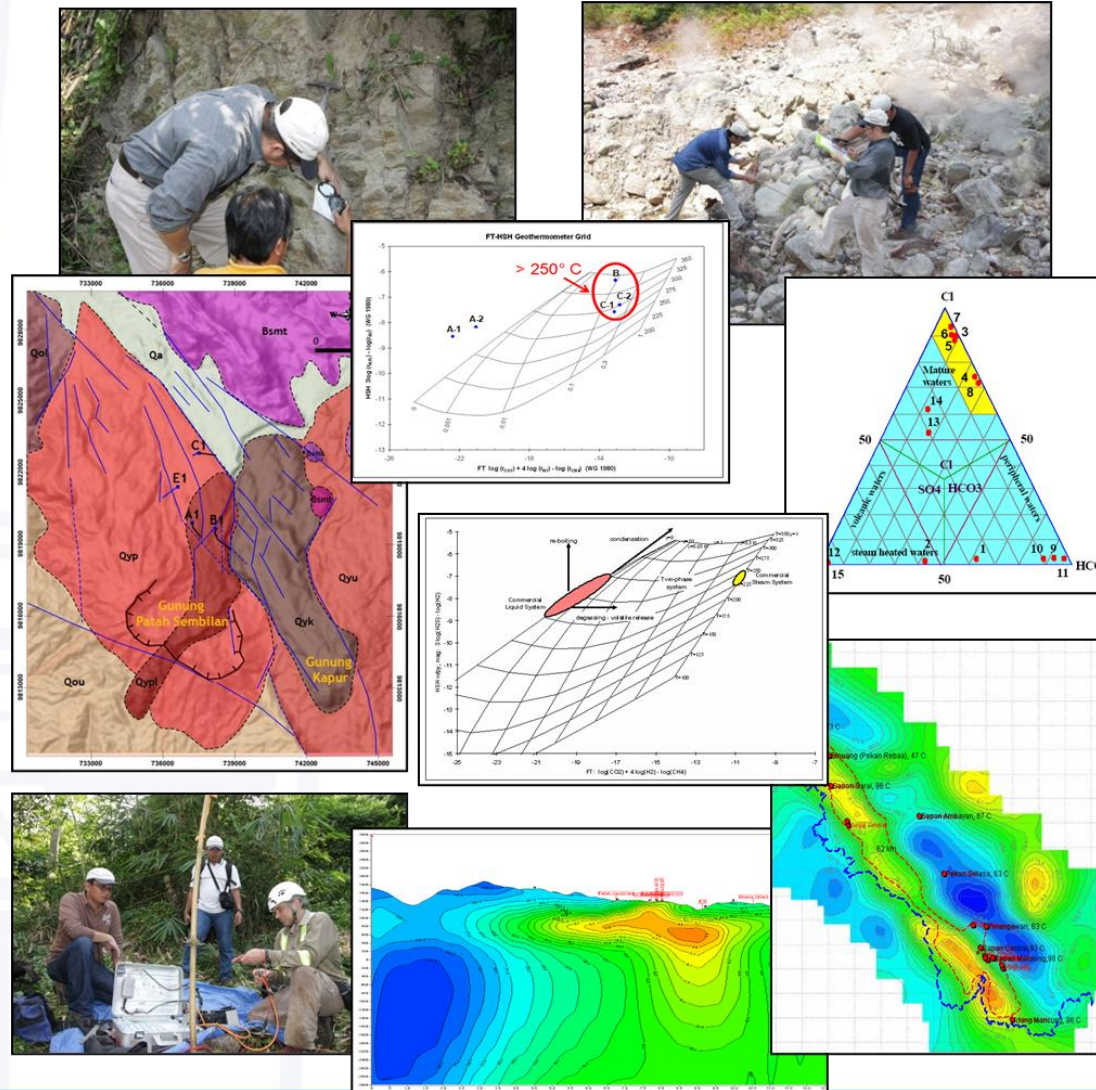
Geothermal “Preferred” System



Muara Laboh well testing

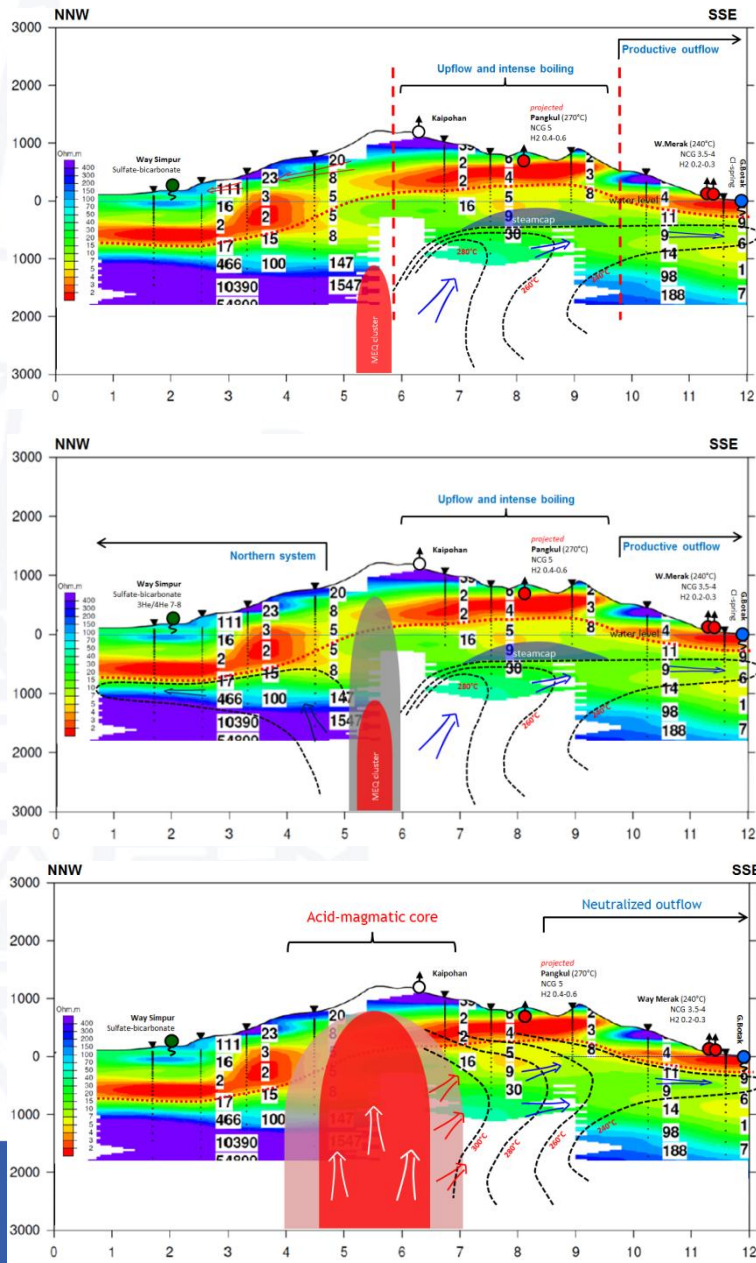
- High Temperature (Enthalpy)
- “Good” Permeability (High productivity)
- Shallow Reservoir Depth
- Sufficient reservoir
- Sustainable resource
- Benign Chemistry

Geothermal exploration



- ✓ Preliminary survey consists of (i) geologic mapping, (ii) geochemistry sampling and analysis, and (iii) geophysical /MT survey
- ✓ Geology provides general framework for conceptual model of geothermal system
- ✓ Geochemistry provides hard data indicates reservoir temp., fluid type and origin, and other data to construct conceptual model
- ✓ Geophysics show subsurface structure which related to reservoir depth, shapes, and size

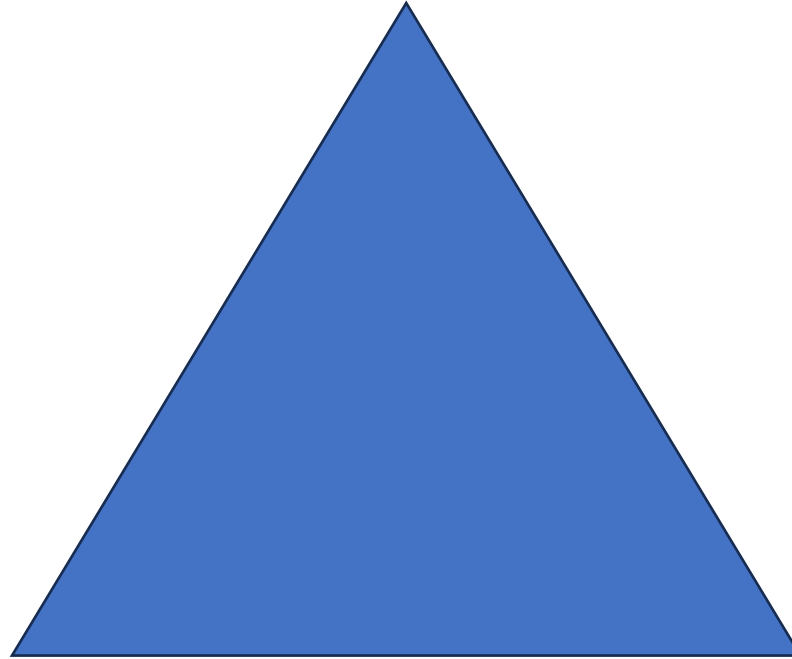
Geothermal exploration (data integration)



- ✓ Integrating geological setting, geochemistry information, geophysical model and thermodynamic principles
- ✓ Geologist leads the process
- ✓ Integrated conceptual model guides further assessments and decisions (resource estimation, exploration drilling strategy, uncertainty management)
- ✓ Alternative models to capture the range of uncertainty

Drilling Objectives??

Company Goals
Economic Viable Project



Drilling Department Goals

- Achieved well objective
- Complete the well according the pre-set design safely
- On schedule (Faster)
- On budget (cheaper)

Subsurface Department Goals

- Drill a productive well
- Test the Conceptual model
- Reduce the reservoir uncertainty

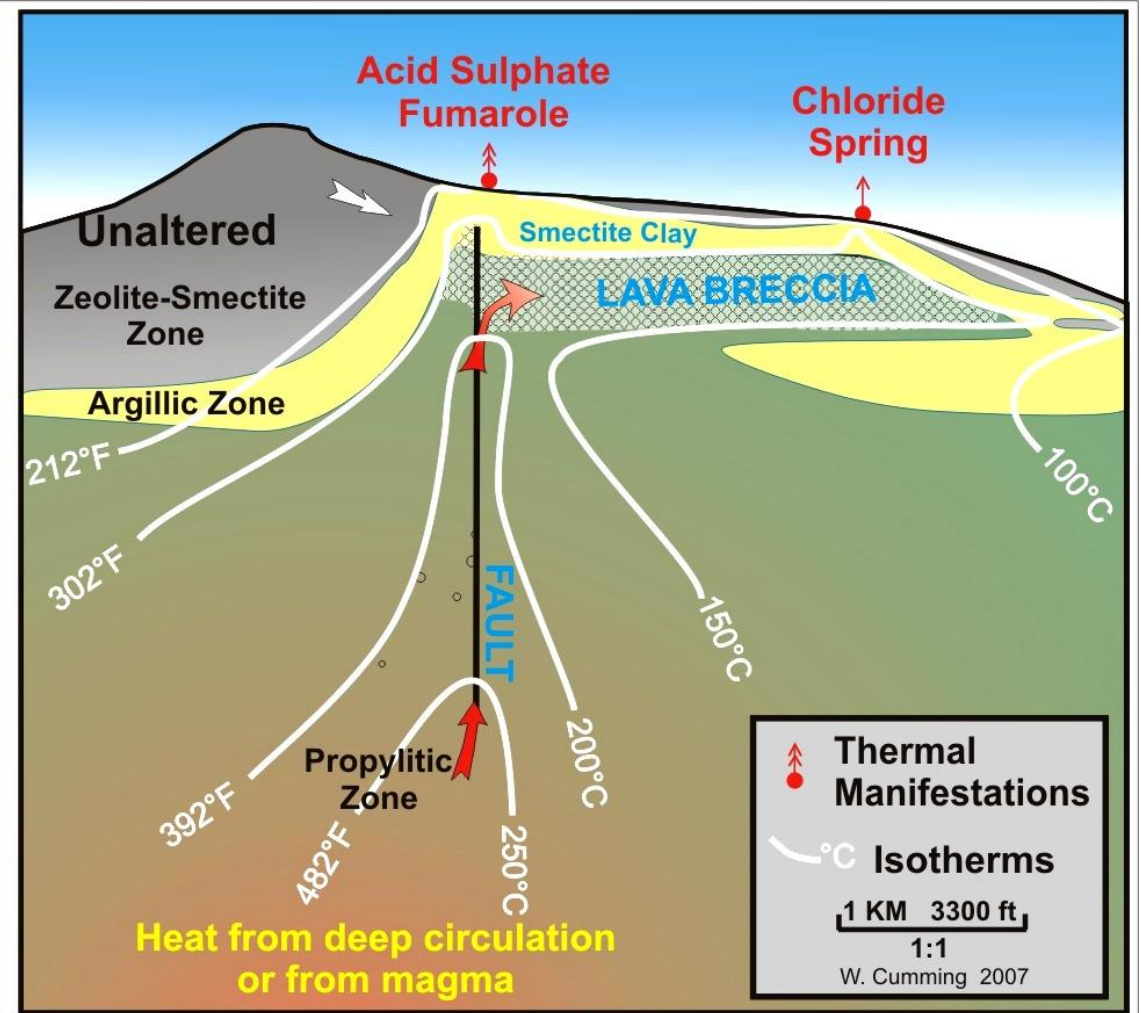
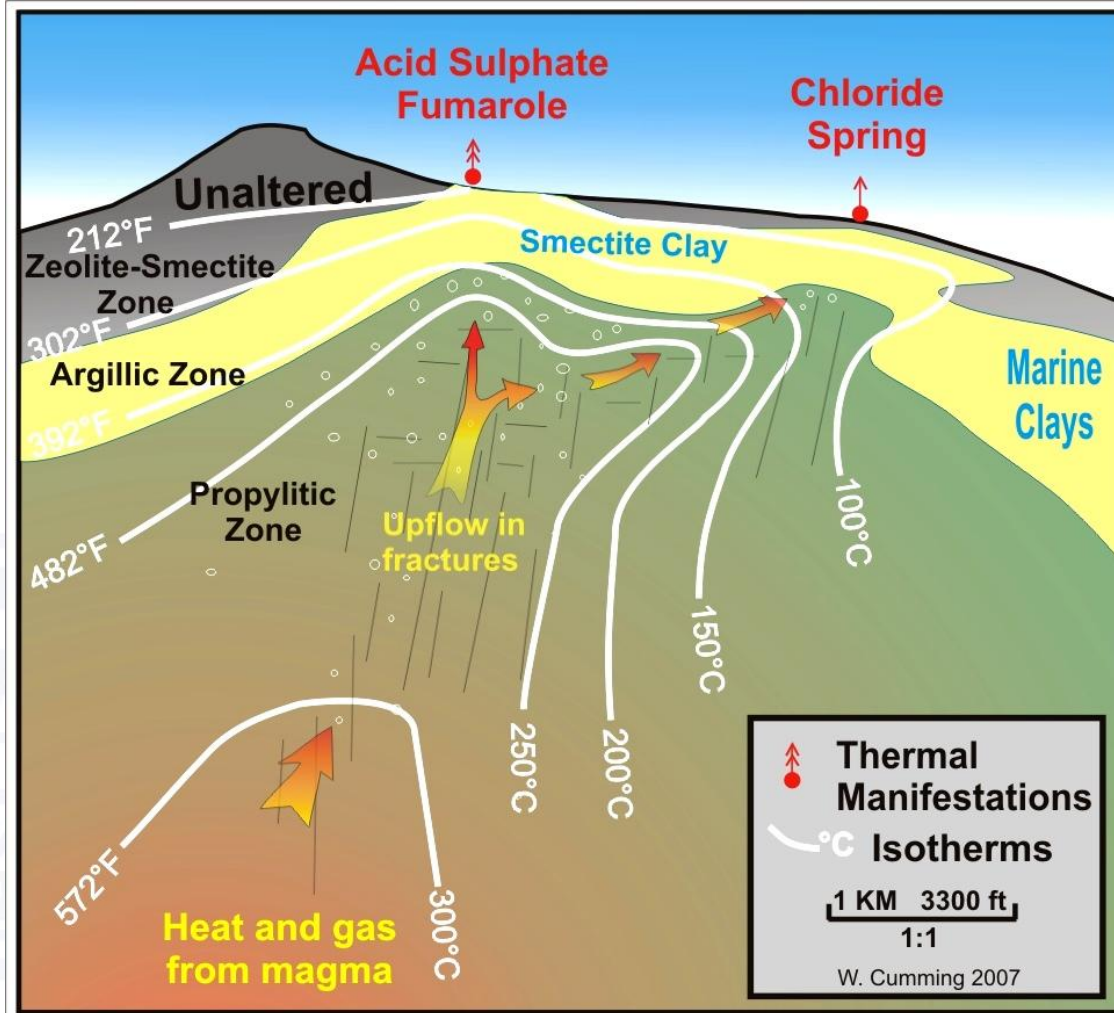
Well Objectives

Objective	Strategy/Tactic
Exploration (Proving Conceptual Model and Resource Area)	Distribute well pads, and move the rig after all the pads its ready high flexibility (e.g. Muara Laboh, Rantau Dedap)
Minimize acid risk	Case deep, avoid shallow fault intersections, use special materials (e.g., Tiwi, Palinpinon, Bac-Man)
Exploit shallow steam cap	Case shallow, use high deviation angle to stay in steam only zone (Salak, Wiyang Windu)
Maximize overall production	Target area between upflow and outflow, with high temperature (>260C) and in region of good structural connection, seek permeable formations. Drill big holes if warranted (Salak)
Forked wells	For deeper targets, and in areas with limited pad potential, drill ML wells (e.g., Salak, Bulalo)
Reduce Costs	In proven areas construct pad and cellars to drill multiple wells from one pad by skidding rig, spacing 250-400 m at ResTop (e.g., Bulalo, Tongonan)
Minimize injection return risk	Find areas downhill from separators with deep ResTop. Case deep, deviate away from production wells >1 km; avoid being along structural grain, or tapping continuous permeable units (e.g., Bulalo)
Optimize injection	Find “safe” distance that sweeps heat and contributes to recharge, but target some injection to be outside
Deep, high T-P productivity	In fields with permeable upflow, case relatively deep to focus on high-enthalpy production (e.g., Olkaria Domes)

- Each well should be thought of as a component of the overall drilling program and its goals and objectives
- Generally a well can only achieve one or two objectives without becoming difficult to implement
- Keep objectives simple and achievable, especially in early exploration
- Must weigh the potential benefits of POS and VOI
 - Some wells might be focused solely on high POS
 - Other wells might have a strong VOI component, with a lower forecast POS

Geothermal system

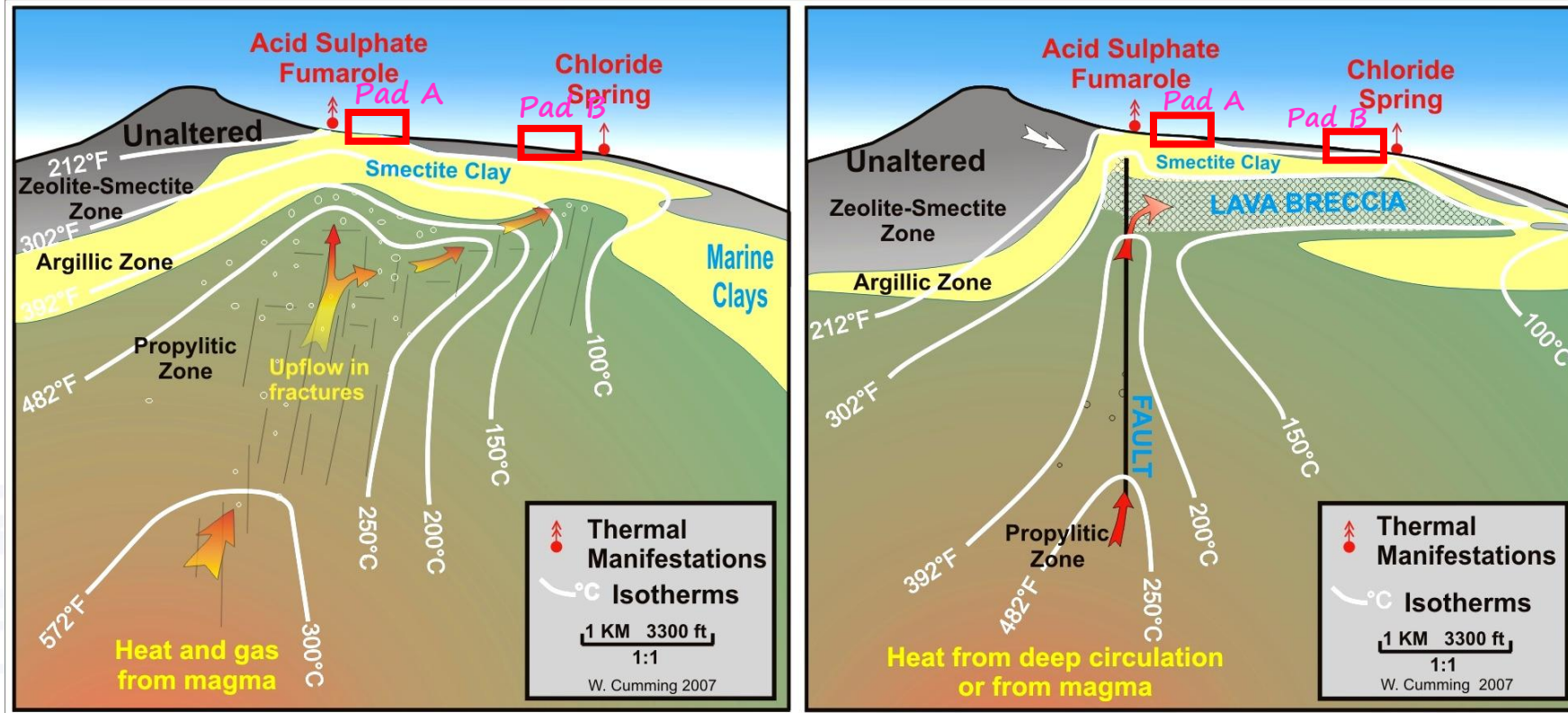
Cumming, 2009



Exploration Drilling Objective : To test the conceptual model

Exercise#1

Cumming, 2009



1. Well Pad Cost (Land, Construction, SWS)
 - Pad A : USD 5 MM
 - Pad B : USD 3 MM
2. Drilling Cost:
 - **Slim Hole (@1000mMD)** : USD 3MM
 - **Standard Hole (@1500mMD)** : USD 5 MM
 - **Big Hole (@2500mMD)** : USD 8 MM
3. Well Testing Cost / well : USD 300 K

A geothermal field located in South America has completed 3Q Exploration and is currently planning to conduct exploration drilling. The approved exploration drilling budget is USD 20 million.

Management has requested that you develop an exploration drilling plan. This plan should include the following details, all within the approved budget:

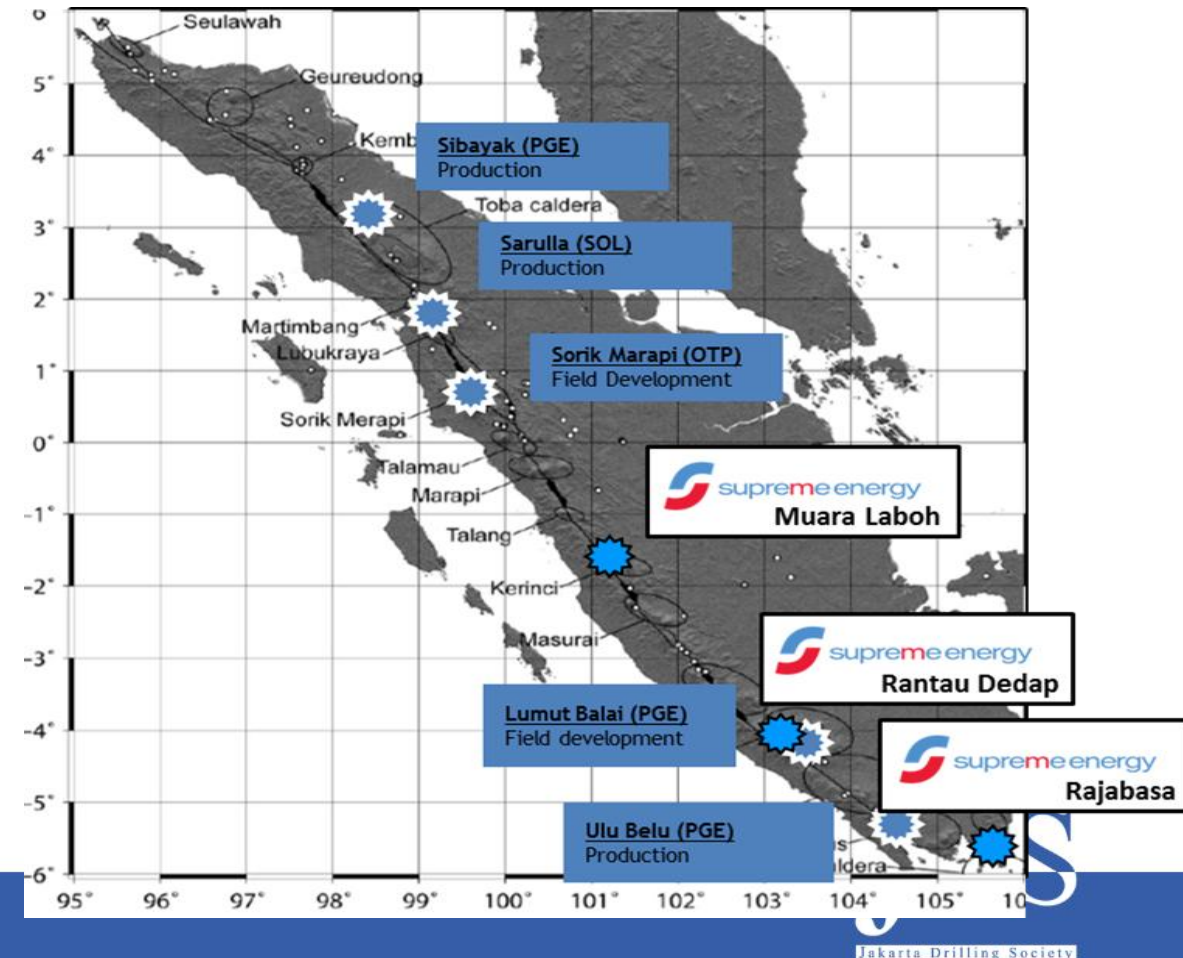
- **Number of Well Pads to Build:** This section should determine the optimal number of well pads to be constructed for the drilling program.
- **Number of Wells to Drill:** Specify the recommended number of wells to be drilled based on the budget and exploration goals.
- **Well Characteristics:**
 - **Type:** Specify the recommended well type (e.g., production well, injection well).
 - **Total Depth:** Define the target total depth for each well.
 - **Inclination:** Define the planned inclination angle for each well.

Please provide an explanation for your recommendations, outlining the rationale behind the chosen number of well pads, wells, and well

Case Studies

Supreme Energy: Three Operation Area

- **Muara Laboh** is located in South Solok Selatan Regency, ~135km SE of Padang, West Sumatera Barat Province
- Rantau Dedap is located ~250km SW of Palembang, South Sumatera Province
- Rajabasa is located in the southern tip of Sumatera Island, about 60 km from Bandar Lampung



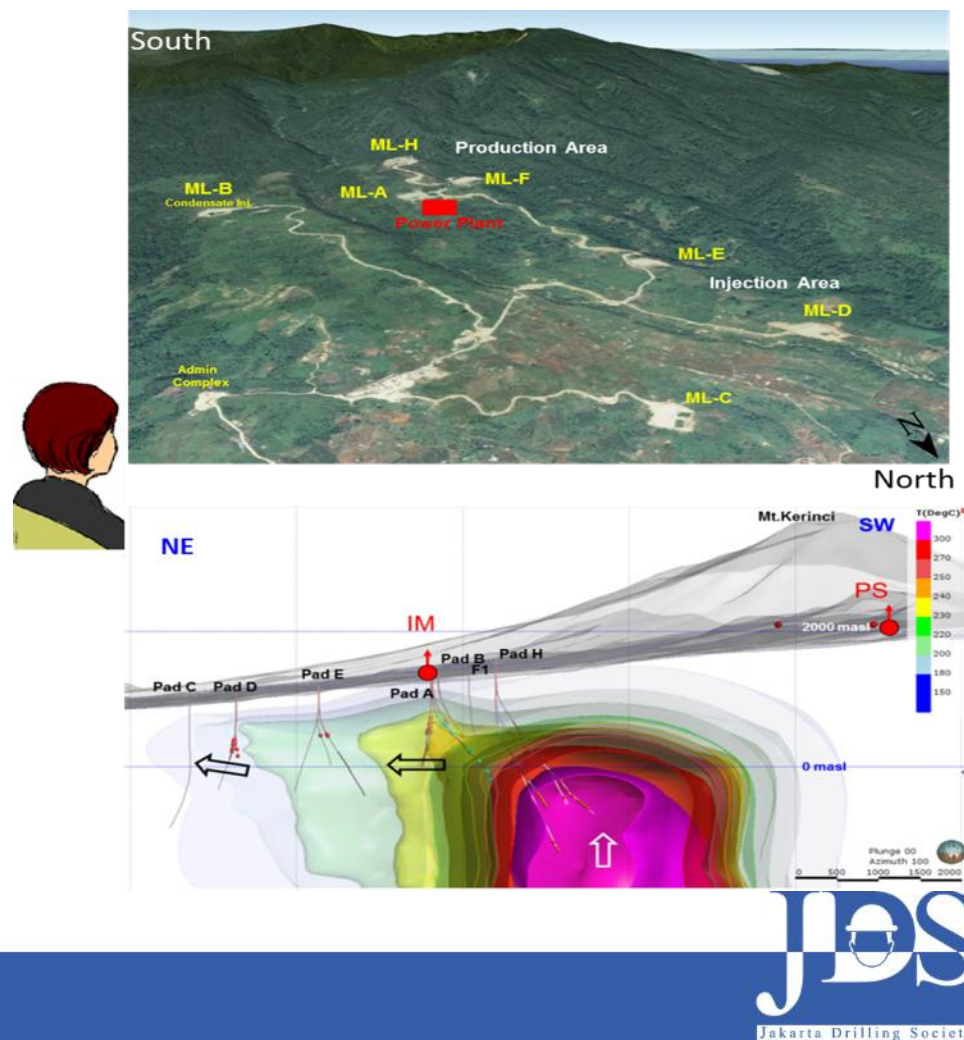
Muara Laboh Geothermal Project Chronology

- 2010: IUP (mining right) granted
- 2012: PPA and BVGL signed
- 2013: Exploration Drilling
- 2014: 60 MWe Resource Confirmation (Single Flash)
- 2015: 80 MWe Development Confirmation (Dual Flash)
- 2016: PPA 2nd amendment
- 2017: Loan agreement signed
- 2018: Development Drilling Completed
- Since December 2019 the field has been on operation at full capacity



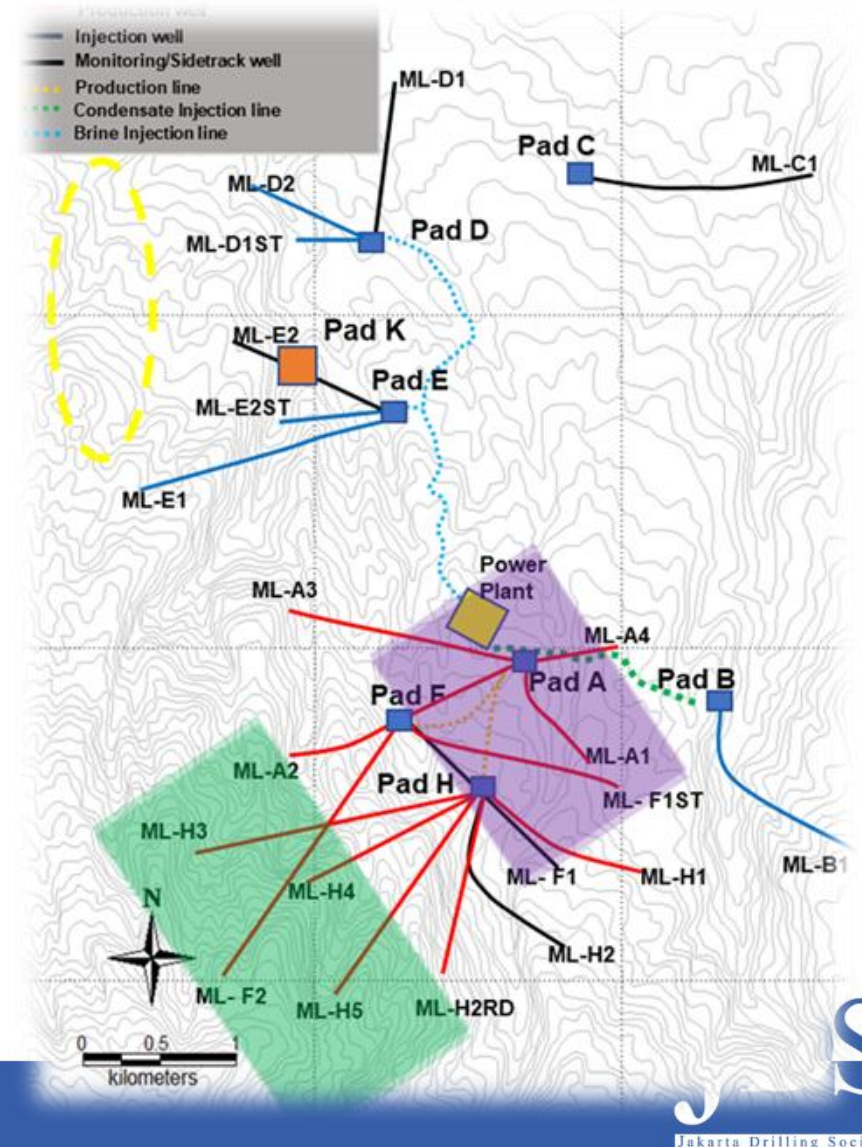
Muara Laboh Resource Development Overview

- Production from high elevation and high temperature area at Pads A, F and H
- Brine injection into lower temperature and lower elevation area at Pads D and E
- Power Plant and Separator Station are close to Pad A
- Condensate injection into Pad B

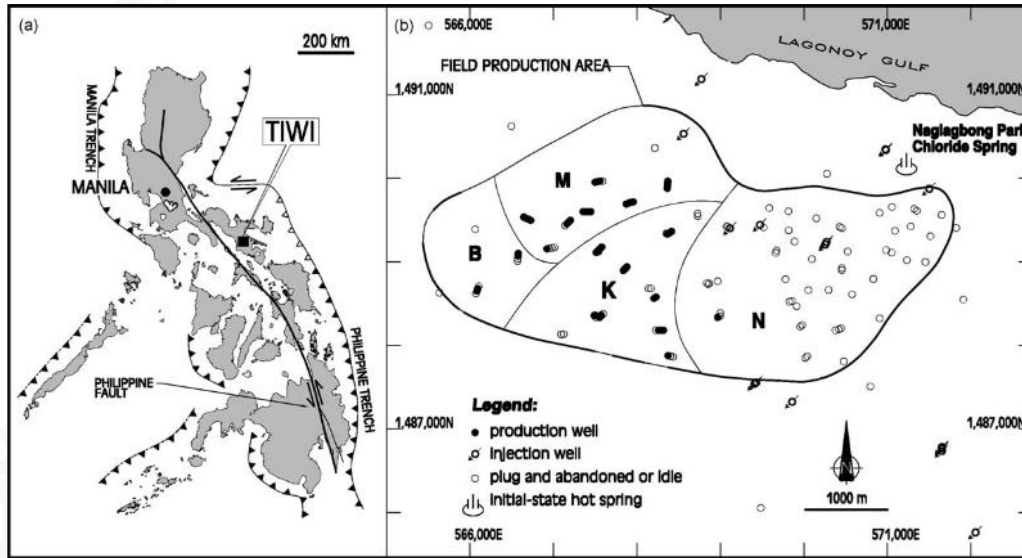


Muara Laboh Stage 2 Resource Development Plan

- Numerical Model Study indicates ML reservoir can support additional ~80MWe
- Production wells are to be drilled deep from the existing Pads F & H (outside National Park) to target SW reservoir underneath the Park
- Separated brine needs to be injected to the western part of the field to provide pressure support to SW reservoir. New Pad K will be constructed
- Injection into Pad E will be limited to minimize premature injection returns to the NE reservoir (Pad A)



Reservoir Management



Tiwi Field
Sugiaman
(2004)

Good Reservoir management:

- ✓ Better understanding on reservoir characteristic and risk management
- ✓ Better control for reservoir management and sustainable field operation
- ✓ Better understanding on the field development potency

